Problem. Find roots of

$$x^3 + a_1 x + a_0 \tag{1}$$

in exact form.
Solution
Case: $a_0, a_1 \in \mathbb{R}$
Attempt a solution of the form

$$x = \sqrt[3]{g} + \sqrt[3]{h}, \tag{2}$$

After inserting x as given in Eq. (2) in Eq. (1) we obtain

$$g + h + 3\sqrt{gh}\left(\sqrt[3]{g} + \sqrt[3]{h}\right) + a_1\left(\sqrt[3]{g} + \sqrt[3]{h}\right) + a_0 = 0 \tag{3}$$

Our goal is to eliminate the radix symbol from the algebra in (3). Therefore, we require:

$$3\sqrt{gh} + a_1 = 0. \tag{4}$$

The requirement in (4) gives

$$g = -\frac{a_1^3}{27h} \tag{5}$$

and

$$g + h + a_0 = 0. \tag{6}$$

After substituting g as given in Eqs. (5) and (6) and rearranging the terms we obtain a quadratic equation in h

$$h^2 + a_0 h - \frac{a_1^3}{27} = 0, \tag{7}$$

which is solved as:

$$h_{1,2} = \frac{-a_0 \pm \sqrt{a_0^2 + \frac{4a_1^3}{27}}}{2}. \tag{8}$$

Together with Eq. (5) (also $g = a_0 - h$) we obtain one set of solutions:

$$h = \frac{-a_0 + \sqrt{a_0^2 + \frac{4a_1^3}{27}}}{2}, \quad g = \frac{-a_0 - \sqrt{a_0^2 + \frac{4a_1^3}{27}}}{2} \tag{9}$$

and

$$x_1 = \sqrt[3]{\frac{-a_0 - \sqrt{a_0^2 + \frac{4a_1^3}{27}}}{2}} + \sqrt[3]{\frac{-a_0 + \sqrt{a_0^2 + \frac{4a_1^3}{27}}}{2}}. \tag{10}$$

One can observe that $x_1 = \sqrt[3]{k_1} + \sqrt[3]{k_2}$, where $k_{1,2}$ are the roots of the equation $k^2 + a_0 k - \frac{a_1^3}{27} = 0$.
Let us denote the discriminant $a_0^2 + \frac{4a_1^3}{27}$ as D.

Now we are left with the task of finding the other two roots x_1 and x_2. This is a simpler task because now we handle a quadratic polynomial, and standard high-school textbooks show how to calculate the roots of such a polynomial.

Given that the polynomial in Eq. (1) can be expressed as

$$(x - x_1)(x - x_2)(x - x_3) = 0 \tag{11}$$

we can work with the equations for the coefficients a_1 and a_0

$$a_1 = x_1 x_2 + x_2 x_3 + x_3 x_1, \tag{12}$$

$$a_0 = -x_1 x_2 x_3. \tag{13}$$

Solving them for x_2 we obtain an equation:

$$x_1^2 x_2^2 - a_0 x_1 + x_3^4 = 0. \tag{14}$$

By solving for x and simplifying we obtain:

$$x_2 = \frac{-x_1 + \sqrt{-3\left(x_1^2 - 4a_1\right)}}{2}. \tag{15}$$

This solution can also be obtained by dividing the original polynomial in (1) with $(x - x_1)$ and finding the roots of the resulting quadratic polynomial.

Since $a_1 = -3\sqrt[3]{gh}$ it can be shown that

$$-3\left(x_1^2 - 4a_1\right) = -3\left(\sqrt{g} - \sqrt{h}\right)^2. \tag{16}$$

Thus we obtain the solutions:

$$x_1 = \sqrt[3]{g} + \sqrt[3]{h} \tag{17}$$

$$x_2 = \frac{-\sqrt[3]{g} - \sqrt[3]{h} + i\sqrt{3}\left(\sqrt[3]{g} - \sqrt[3]{h}\right)}{2} \tag{18}$$

$$x_3 = \frac{-\sqrt[3]{g} - \sqrt[3]{h} - i\sqrt{3}\left(\sqrt[3]{g} - \sqrt[3]{h}\right)}{2}, \tag{19}$$

or

$$x_1 = \sqrt[3]{g} + \sqrt[3]{h} \tag{20}$$

$$x_2 = \frac{\sqrt[3]{h}\left(-1 - \sqrt{3}i\right) + \sqrt[3]{g}\left(-1 + \sqrt{3}i\right)}{2} \tag{21}$$

$$x_3 = \frac{\sqrt[3]{h}\left(-1 + \sqrt{3}i\right) + \sqrt[3]{g}\left(-1 - \sqrt{3}i\right)}{2}. \tag{22}$$

The solution becomes more illuminating after we recognise that $-1 + \sqrt{3}i = e^{\frac{2\pi i}{3}}$ and $-1 - \sqrt{3}i = e^{\frac{4\pi i}{3}}$, see Fig. 1. Furthermore, $e^{\frac{4\pi i}{3}} = e^{-\frac{2\pi i}{3}}$. Hence the roots can be written in the form shown in Eqs (23), (24) and (25):

For a_0 and $D \in \mathbb{R}$ and $D \geq 0$:

$$\boxed{\begin{aligned} x_1 &= \sqrt[3]{g} + \sqrt[3]{h} && (23)\\ x_2 &= \sqrt[3]{h}e^{\frac{-2\pi i}{3}} + \sqrt[3]{g}e^{\frac{2\pi i}{3}} && (24)\\ x_3 &= \sqrt[3]{h}e^{\frac{2\pi i}{3}} + \sqrt[3]{g}e^{-\frac{2\pi i}{3}} && (25)\end{aligned}}$$

where

$$h = \frac{-a_0 + \sqrt{D}}{2}, \quad g = \frac{-a_0 - \sqrt{D}}{2} \tag{26}$$

and

$$D = a_0^2 + \frac{4a_1^3}{27}. \tag{27}$$

In the case $a_0 \in \mathbb{R}$ and $D = 0$ there are two equal roots:

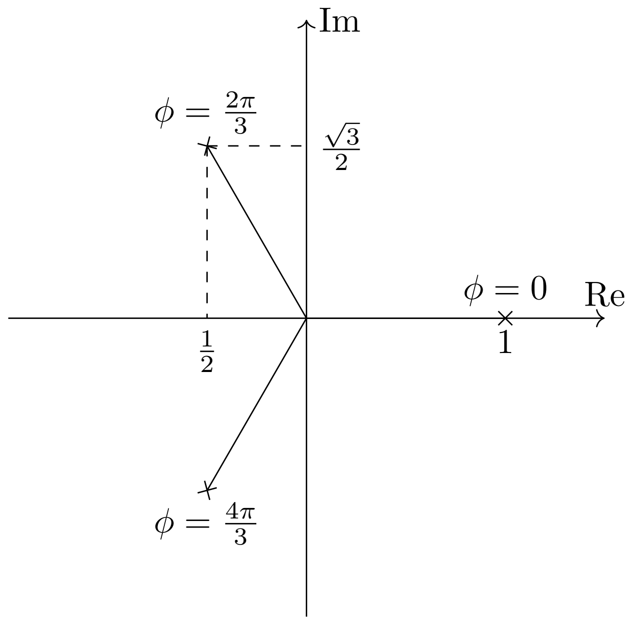

Figure 1: e^{i0}, $e^{\frac{2\pi i}{3}}$, $e^{\frac{4\pi i}{3}}$ plotted on the complex plane

For $a_0 \in \mathbb{R}$ and $D = 0$:

$$\boxed{g = h = -\frac{a_0}{2}.} \tag{28}$$

$$x_1 = -2\sqrt[3]{\frac{a_0}{2}} \tag{29}$$

$$x_2 = x_3 = \sqrt[3]{\frac{a_0}{2}}. \tag{30}$$

For the case $a_0, D_0 \in \mathbb{R}$ and $D < 0$ we observe that:

$$|g| = |h| = \frac{-a_1}{3}, \tag{31}$$

$$\sqrt[3]{h} = \sqrt[2]{\frac{-a_1}{3}} e^{\frac{i\phi_h}{3}}, \tag{32}$$

$$\sqrt[3]{g} = \sqrt[2]{\frac{-a_1}{3}} e^{\frac{-i\phi_h}{3}}, \tag{33}$$

$$\sqrt[3]{h} + \sqrt[3]{g} = 2\sqrt{\frac{-a_1}{3}} \cos\left(\frac{\phi_h}{3}\right), \tag{34}$$

$$\sqrt[3]{h} - \sqrt[3]{g} = 2i\sqrt{\frac{-a_1}{3}} \sin\left(\frac{\phi_h}{3}\right). \tag{35}$$

where ϕ_h is the phase of h (see (72)).

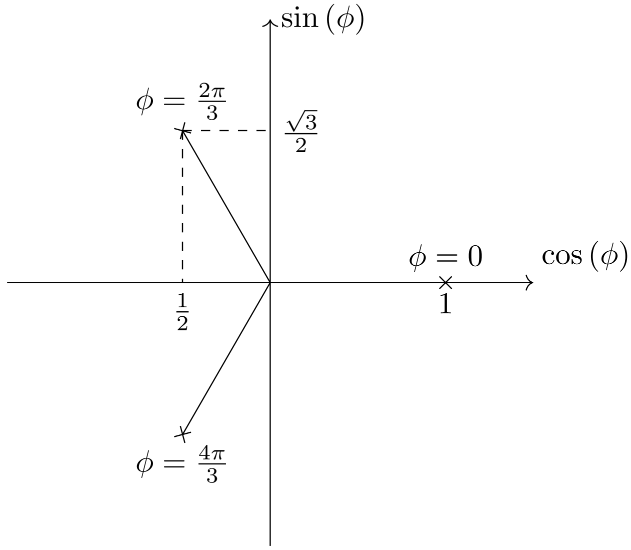

Figure 2: $\phi = 0$, $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ plotted in polar coordinates

Therefore,

$$x_1 = 2\sqrt{\frac{-a_1}{3}}\cos\left(\frac{\phi_h}{3}\right), \tag{36}$$

$$x_2 = \sqrt{\frac{-a_1}{3}}\left(-\cos\left(\frac{\phi_h}{3}\right) + \sqrt{3}\sin\left(\frac{\phi_h}{3}\right)\right), \tag{37}$$

$$x_3 = \sqrt{\frac{-a_1}{3}}\left(-\cos\left(\frac{\phi_h}{3}\right) - \sqrt{3}\sin\left(\frac{\phi_h}{3}\right)\right). \tag{38}$$

By noting that (see Fig. 2)

$$\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} \qquad \sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2} \tag{39}$$

$$\cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2} \qquad \sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2} \tag{40}$$

x_2 and x_3 can be expressed as

$$x_2 = 2\sqrt{\frac{-a_1}{3}}\left(\cos\left(\frac{2\pi}{3}\right)\cos\left(\frac{\phi_h}{3}\right) + \sin\left(\frac{2\pi}{3}\right)\sin\left(\frac{\phi_h}{3}\right)\right), \tag{41}$$

$$x_3 = 2\sqrt{\frac{-a_1}{3}}\left(\cos\left(\frac{4\pi}{3}\right)\cos\left(\frac{\phi_h}{3}\right) + \sin\left(\frac{4\pi}{3}\right)\sin\left(\frac{\phi_h}{3}\right)\right). \tag{42}$$

By using the relationship

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) \tag{43}$$

the roots can be recast in a more illuminating form as given in equations (44), (45), (46).

For a_0 and $D \in \mathbb{R}$ and $D < 0$:

$$\boxed{\begin{aligned} x_1 &= 2\sqrt{-\frac{a_1}{3}}\cos\left(\frac{\phi_h}{3}\right), \\ x_2 &= 2\sqrt{-\frac{a_1}{3}}\cos\left(\frac{\phi_h}{3} - \frac{2\pi}{3}\right), \\ x_3 &= 2\sqrt{-\frac{a_1}{3}}\cos\left(\frac{\phi_h}{3} + \frac{2\pi}{3}\right). \end{aligned}} \quad \begin{aligned} (44) \\ (45) \\ (46) \end{aligned}$$

where

$$h = \frac{-a_0 + i\sqrt{-D}}{2}, \quad g = \frac{-a_0 - i\sqrt{-D}}{2} \tag{47}$$

and

$$D = a_0^2 + \frac{4a_1^3}{27}, \tag{48}$$

and the phase ϕ_h is taken with respect to h and is calculated as

$$\phi_h = \arctan\left(\frac{\sqrt{-D}}{-a_0}\right) + n\pi, \quad \text{where } n = 0 \quad \text{if } a_0 < 0, \quad n = 1 \quad \text{if } a_0 > 0 \tag{49}$$

$$= \frac{\pi}{2}, \quad \text{if } a_0 = 0 \tag{50}$$

From equations (44), (45), (46) it can be seen that the roots follow each other by phase $\frac{2\pi}{3}$ in the sequence x_2—x_1—x_3.

Eqs. (44), (45), (46) are generators of interesting identities. For example, trying the roots $-1, -2, 3$, we find that:

$$3 = 2\sqrt{\frac{7}{3}}\cos\left(\frac{1}{3}\arctan\left(\frac{10}{9\sqrt{3}}\right)\right) \tag{51}$$

Now, we would like to develop the general solution for the case $a_0, a_1 \in \mathbb{C}$, $a_1 \neq 0$. We consider g and h as complex numbers:

$$g = |g|e^{(i(\phi_g + n_g 2\pi))}, \tag{52}$$

$$h = |h|e^{(i(\phi_h + n_h 2\pi))}, \tag{53}$$

$$\tag{54}$$

where n_g are n_h integers. While g and h have equal values for all n_g and n_h respectively, the roots of g and h don't as we will see shortly. From the requirement (4) it follows that

$$3\sqrt[3]{|g||h|}e^{\frac{i(\phi_g + \phi_h + (n_g + n_h)2\pi)}{3}} = |a_1|e^{i(\phi_a + \pi)}. \tag{55}$$

Hence

$$n_h + n_g = \frac{3(\phi_a + \pi) - \phi_g - \phi_h}{2\pi}. \tag{56}$$

We note the gauge freedom $-n_h$ and n_g can be any real integer, only their sum must satisfy Eq. (56).

After cubing both sides of the equation (4) we obtain:

$$|g|e^{i\phi_g} = -\frac{a_1^3}{27|h|e^{i\phi_h}} \tag{57}$$

and

$$\left(|h|e^{i\phi_h}\right)^2 + a_0|h|e^{i\phi_h} + a_0 = 0. \tag{58}$$

We obtain the g and h solutions

$$|h|e^{i\phi_h} = \frac{-a_0 + \sqrt{D}}{2}, \tag{59}$$

$$|g|e^{i\phi_h} = \frac{-a_0 - \sqrt{D}}{2} \tag{60}$$

From (17), (18) and (19) we obtain

$$x_1 = \sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3}} + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3}} \tag{61}$$

$$x_2 = \frac{\sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3}}\left(-1 - \sqrt{3}i\right) + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3}}\left(-1 + \sqrt{3}i\right)}{2} \tag{62}$$

$$x_3 = \frac{\sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3}}\left(-1 + \sqrt{3}i\right) + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3}}\left(-1 - \sqrt{3}i\right)}{2} \tag{63}$$

After simplifying, we obtain the following:

For a_0 and $D \in \mathbb{C}$ and $a_1 \neq 0$:

$$\boxed{\begin{aligned} x_1 &= \sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3}} + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3}}, & (64)\\ x_2 &= \sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3} - \frac{2\pi i}{3}} + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3} + \frac{2\pi i}{3}}, & (65)\\ x_3 &= \sqrt[3]{|h|}e^{\frac{i(\phi_h + n_h 2\pi)}{3} + \frac{2\pi i}{3}} + \sqrt[3]{|g|}e^{\frac{i(\phi_g + n_g 2\pi)}{3} - \frac{2\pi i}{3}}, & (66) \end{aligned}}$$

where

$$h = \frac{-a_0 + \sqrt{D}}{2}, \quad g = \frac{-a_0 - \sqrt{D}}{2} \tag{67}$$

and

$$D = a_0^2 + \frac{4a_1^3}{27}. \tag{68}$$

$$n_h + n_g = \frac{3\left(\phi_a + \pi\right) - \phi_g - \phi_h}{2\pi} \tag{69}$$

and the phase

$$\begin{aligned} \phi_f &= \arctan\left(\frac{i\left(f^* - f\right)}{f + f^*}\right) + n\pi, \quad \text{where } n = 0 \quad \text{if } f + f^* > 0, \quad n = 1 \quad \text{if } f + f^* < 0 & (70)\\ &= \frac{\pi}{2}, \quad \text{if } a_0 = 0 \quad \text{and} \quad i\left(f^* - f\right) > 0, & (71)\\ &= \frac{3\pi}{2}, \quad \text{if } a_0 = 0 \quad \text{and} \quad i\left(f^* - f\right) < 0, & (72) \end{aligned}$$

where f^* denotes the complex conjugate of f.

The general solution is depicted in tgee Argand diagram in Figure 3, where we have introduced new variables for convenience:

$$H = \sqrt[3]{h}e^{i\left(\frac{\phi_h}{3} + \frac{n_h 2\pi}{3}\right)}, \tag{73}$$

$$G = \sqrt[3]{g}e^{i\left(\frac{\phi_g}{3} + \frac{n_g 2\pi}{3}\right)}. \tag{74}$$

In the case of $a_1 = 0$ we cannot use the solution for $a_1 \neq 0$ because the phase of a_1 is not defined. In this case $h = 0$ and $g = |a_0|e^{\pi i}$. By plugging in these values in Eqs. (23), (24) and (25) we obtain the following roots:

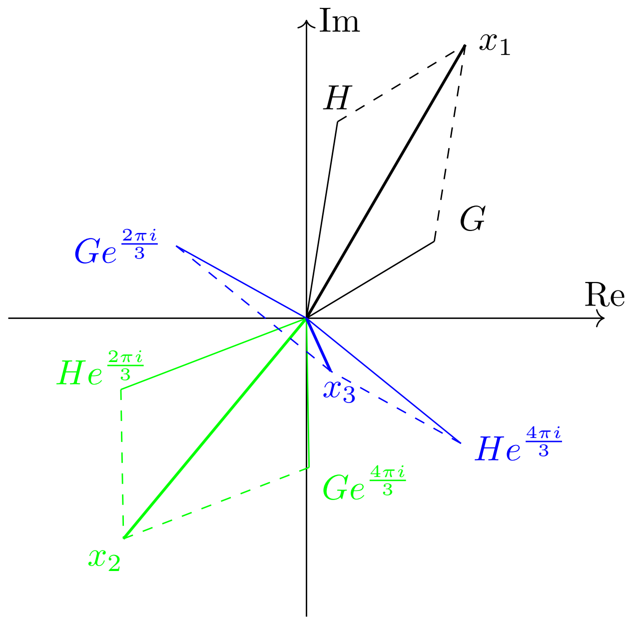

Figure 3: Argand diagram of a general solution

For a_0 and $D \in \mathbb{R}$ and $D \geq 0$:

$$\boxed{\begin{aligned} x_1 &= \sqrt[3]{|a_0|}e^{\frac{(\phi_{a_0}+\pi)i}{3}}, \\ x_2 &= \sqrt[3]{|a_0|}e^{\frac{(\phi_{a_0}+5\pi)i}{3}}, \\ x_3 &= -\sqrt[3]{|a_0|}e^{\frac{\phi_{a_0} i}{3}}. \end{aligned}} \qquad \begin{aligned} (75) \\ (76) \\ (77) \end{aligned}$$

In the case $a_0 = 0$, $\sqrt[3]{g} = -\sqrt{\frac{a_1}{3}}$, $\sqrt[3]{h} = \sqrt{\frac{a_1}{3}}$ the following solutions are obtained:

If $a_0 = 0$:

$$\boxed{\begin{aligned} x_1 &= 0, \\ x_2 &= i\sqrt{|a_1|}e^{i\phi_{a_1}} \\ x_3 &= -i\sqrt{|a_1|}e^{i\phi_{a_1}}. \end{aligned}} \qquad \begin{aligned} (78) \\ (79) \\ (80) \end{aligned}$$

General case

We want to find the solutions of the equation

$$b_3x^3 + b_2x^2 + b_1x + b_0 = 0. \tag{81}$$

We introduce the substitution $x = x' + p$ and obtain:

$$b_3x'^3 + (3b_3p + b_2)\,x'^2 + \left(3b^3p^2 + 2pb_2 + b_1\right)x' + b_3p^3 + b_2p^2 + b_1p + b_0 = 0. \tag{82}$$

We require the $x^{'2}$ term to vanish, hence:

$$p = -\frac{b_2}{3b_3}. \tag{83}$$

From Eqs. (82) and (83) we obtain the coefficients a_0 and a_1 in Eq. (1):

$$a_1 = \frac{1}{b_3}\left(-\frac{b_2^3}{3b_3} + b_1\right), \tag{84}$$

$$a_0 = \frac{1}{b_3}\left(\frac{b_2^3}{27b_3^2} - \frac{b_1 b_2}{3b_3} + b_0\right). \tag{85}$$

Subsequently, we find the roots of the polynomial of x' in Eq. (1), and afterwards transform $x = x' + p$.